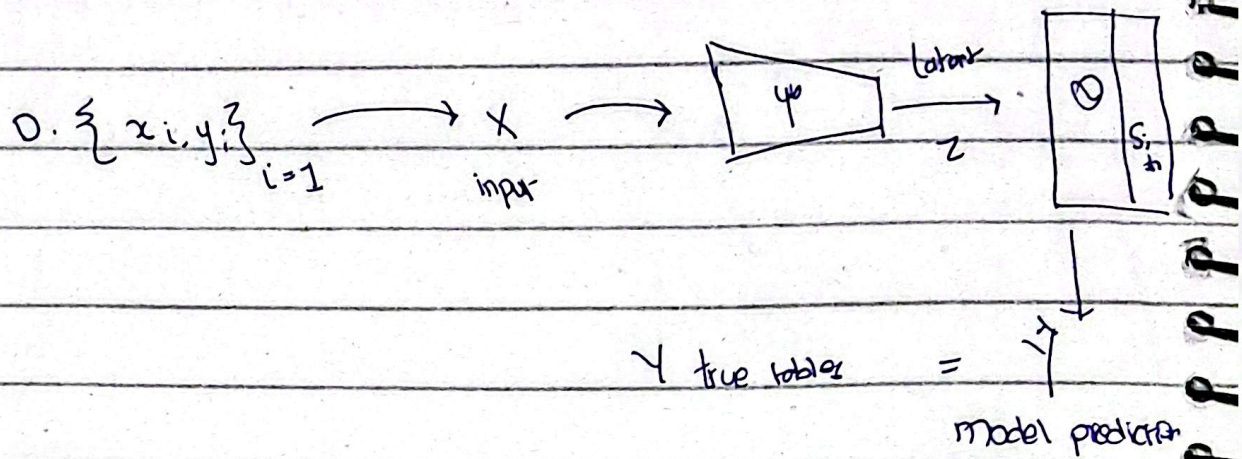
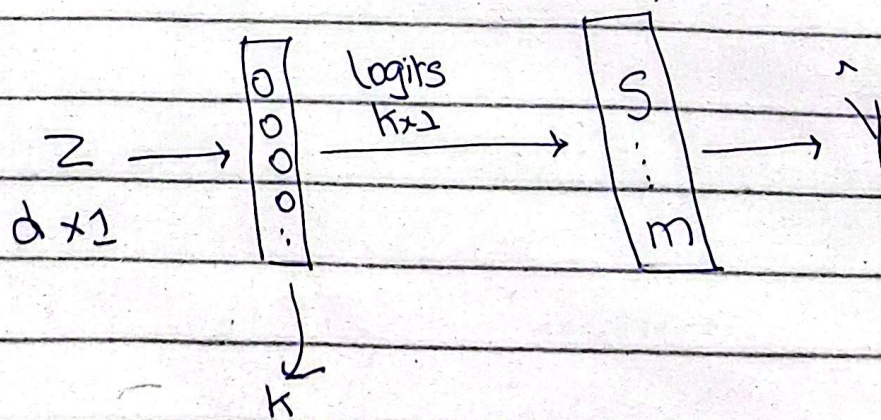
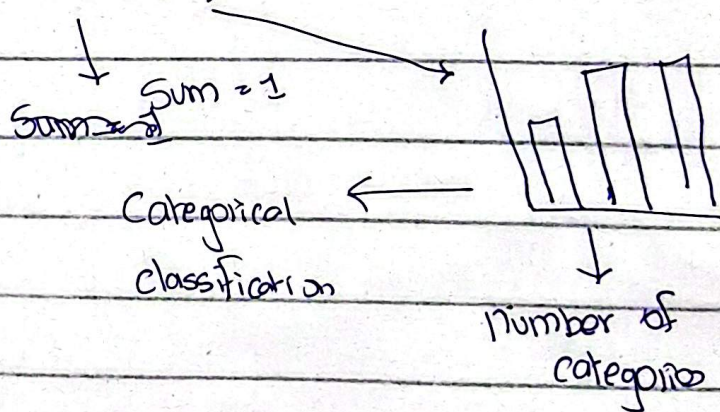


Lecture 2



→ Distribution:

$$P(X), P(Z), P(Y), P(X, Y), P(Y|X), \dots, P(Y|X) \dots$$



Class entropy:

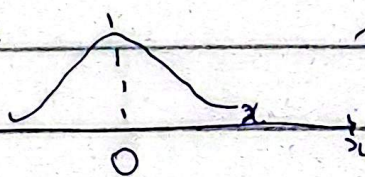
basically maximising dot product for
correct prediction

minimising dot product with other classes

$$L_{CE} = -\log \hat{y}_j \quad \text{if true category for } x=j$$

→ X is a random variable having distribution $P(x)$

→ The most common random variable is gaussian
Random variable



$$P(x) \sim N(0, 1)$$

→ mean

→ standard variance

→ SD is higher dimension is covariance

if $x \sim P(x) = N(0, I)$

$\downarrow$ $D \times 1$ $\downarrow$ $D \times 1$ $\downarrow$ $D \times D$

has no dependency with
other variables

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Covariance is basically each dimension has information about itself and with itself and other dimensions

if $I \approx [3 \times 3] \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow$ no dependency with others

independent dependency $\rightarrow$ independent
 $\downarrow$ random variable

each dimension has information about itself only.

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

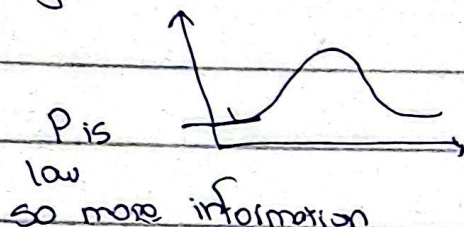
basically still independent, no cross talk

⇒ information content:

Random quantity of a random variable

→ anything that has probability of occurrence low has a lot of information inside it

like in gaussian

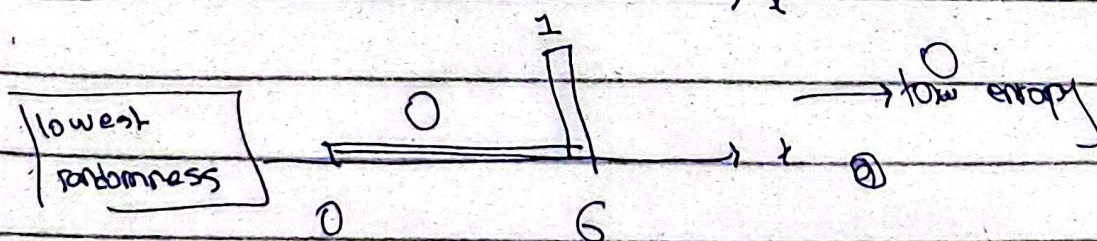
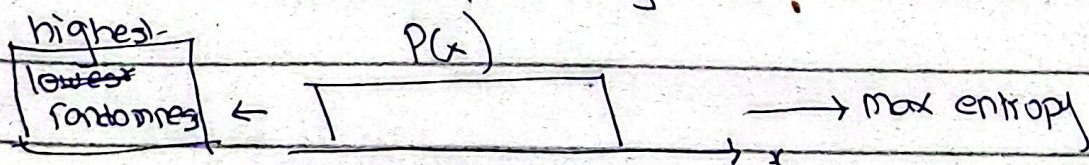


$$\text{Information content} = -\log_2 (P(x = x_k))$$

⇒ Average information [entropy]:

$$H(x) = -\sum_x P(x) \log_2 P(x)$$

- each point's probability & information in it.



Randomness & Entropy

Entropy is the measure of randomness.

→ x 's entropy will be highest

→ $H(y|x)$ will be lower as the model's confidence grows

$$\Rightarrow P(x) \sim N(\mu, R)$$

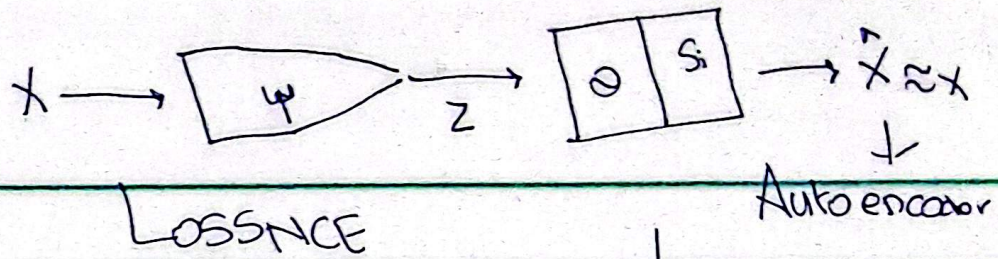
↓
multi-dimensional gaussian

Explain on its own

Focus on shape and properties

Loss Function

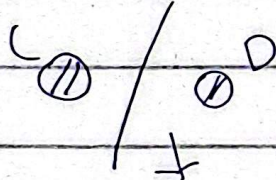
— Prediction and minimize divergence between the two



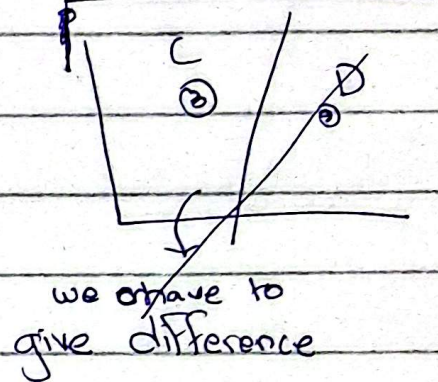
1. Behavior / Geometry / Shape.

MSE ($x, \hat{x}$)

Reconstruction loss

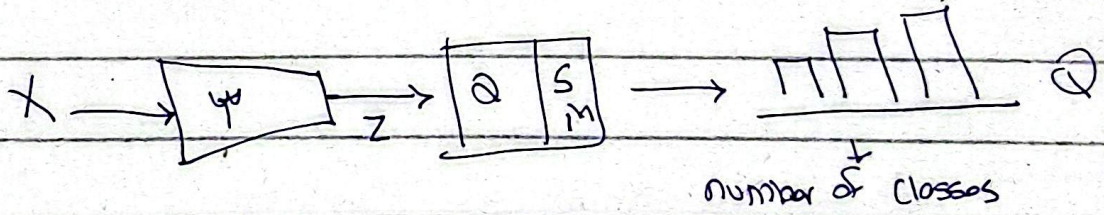


basically making sure C and D have difference between them, so model is confused, So basically we are changing geometry of the representations

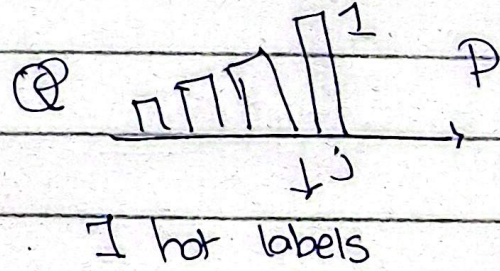


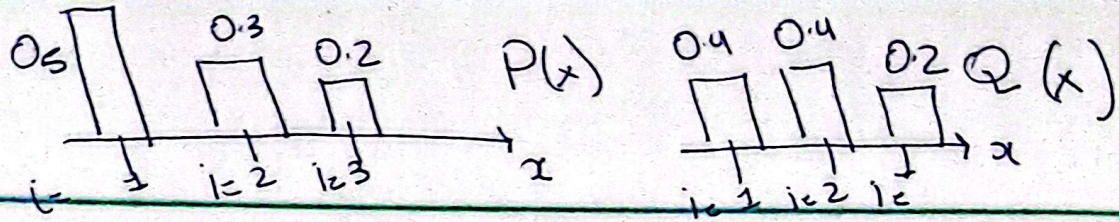
2. Distribution - Reference loss.

Categorical Classification



making sure Q is shaped into to look like same as P





1. Geometry

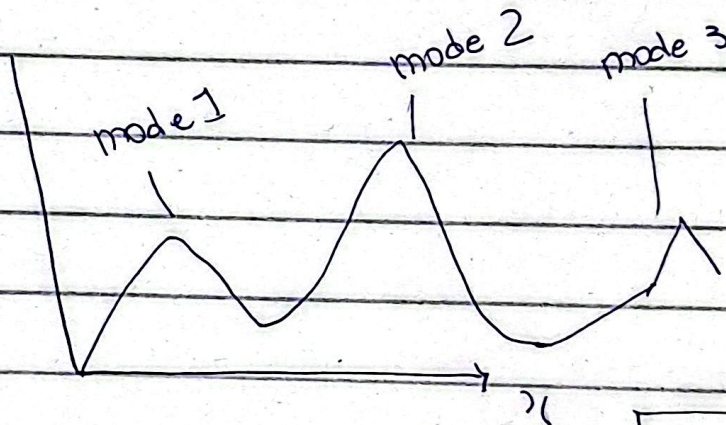
↳ location of x lobes

⇒ assumption that $P(x)$ is overlapping each other spot

2. Probabilities mass

↓
axis

↑
Prob mass



1, 2, 3
are states

1. Distance — minimise

F - divergence:

$\frac{P}{Q}$ or $\frac{Q}{P}$
Ratio

Take ratio of each
state and find
distance

→ Does not know $P(x)$

→ So expectation

Expectation: $E(x)$

$$E(x) = \int x \cdot P(x) dx$$

Continuous

- Forward ~~KZ~~ divergence.

→ Statistics

⇒ basically we don't know $P(x)$

to find $E(x)$, so we

take expectation at each sample

$$E[g(x)] = \int g(x) P(x) dx$$

↓
Statistical average

~~$\frac{1}{N} \sum_{i=1}^N g(x_i)$~~

$$\frac{1}{N} \sum_{i=1}^N g(x_i)$$

Sample Average

Expected value of anything

$$E[AT] = \int AT \cdot P(x) dx$$

$$\sum x_i (\Delta x) P(x_i)$$

e.g.

$$E[g(x)] = \int g(x) P(x) dx$$

Anything will always be function of that random

As N increases, Sample average

becomes equal to Statistical average

↓

$\frac{1}{N} \rightarrow$ Each sample's probability of happening is 1

Support (mode) Seeking

$$D_{KL}^F(P||Q) = \sum_i P_i \log \left(\frac{P_i}{Q_i} \right)$$

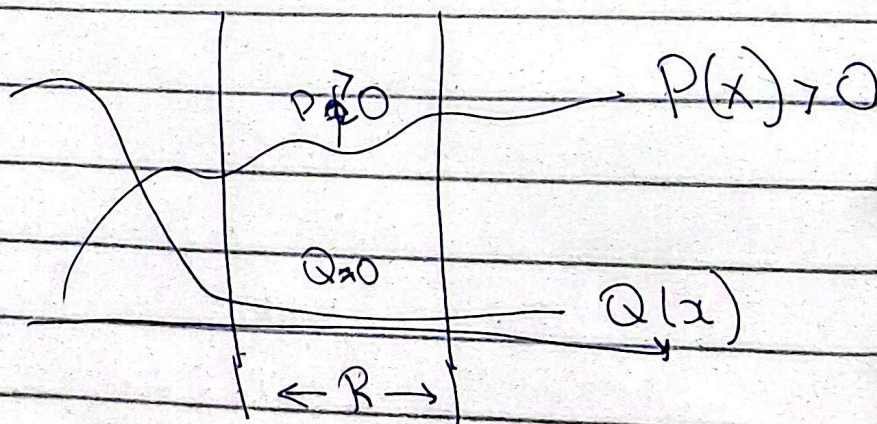
$$P-Q = (0.1, -0.1, 0)$$

$$P/Q = (1.025, 0.75, 1)$$

P has ~~relat~~ 0.25 probability ^{higher} relative to Q

→ From example on last page 0.025061

Calculus on my own



$$D_{KL}^F \rightarrow \inf_{\infty}$$

$\textcircled{Q} \rightarrow \text{Inf}$

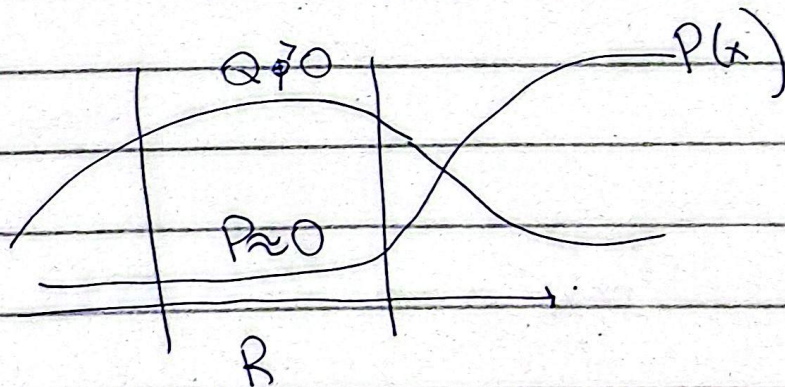
Sport-Speaking

→ basically in this region R, mass of Q is less than $\textcircled{P}$, so that region R's $\textcircled{Q}$ support are penalized:

↓

meaning Q has to add mass to P

$$D_{KL R}(Q||P) \triangleq \sum_{i=1}^3 Q_i \log\left(\frac{Q_i}{P_i}\right)$$



↓

$D_{KL R}$ will below to infinity ∞

↓ Penalize

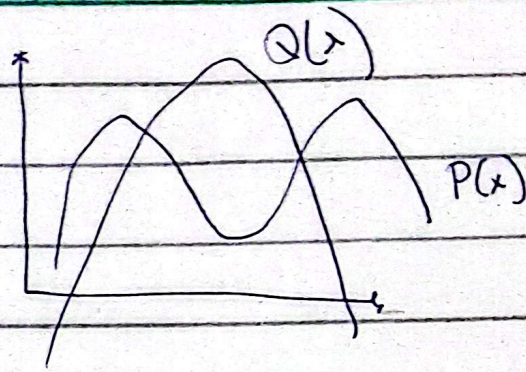
Q will start to decrease mass

and will start to to the position where

$Q < P$, so in R mass of P mass of Q

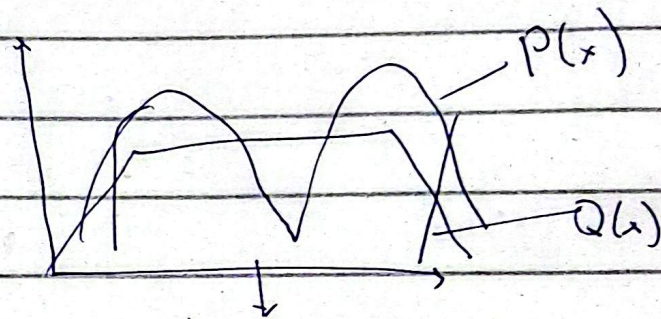
Mode covering

Mode covering =



Q move to left/right

Mode Seeking



Q moves to average

TVD = Total variation distance

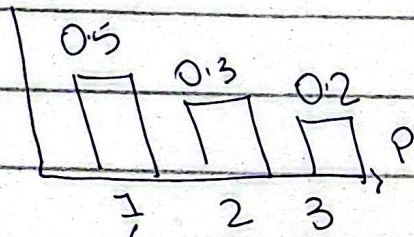
$$P-Q \rightarrow TVD = \frac{1}{2} \sum_i |P_i - Q_i|$$

$$MMD^2 : MMD^2 = (P-Q)^T K (P-Q)$$

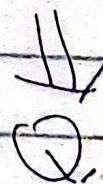
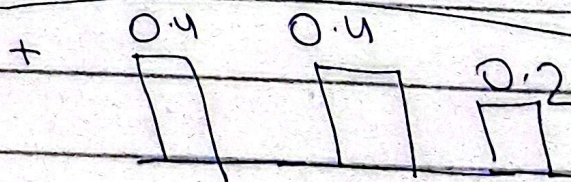
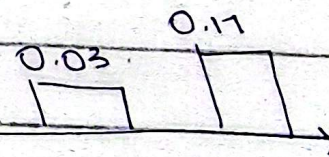
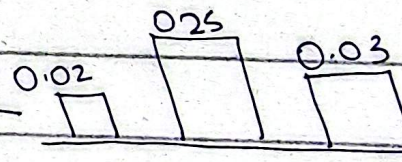
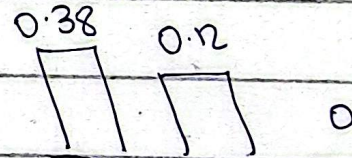
Mass Mean
Discrepancy

Optimal Transport

$$\pi^* = \begin{bmatrix} 0.38 & 0.12 & 0 \\ 0.02 & 0.25 & 0.03 \\ 0 & 0.03 & 0.17 \end{bmatrix}$$



distribute acc to 1st row of π^*



→ optimal transport give a way to ~~π~~ ^{*}
 $P \rightarrow Q$

For Q to P , we take
 π^*

* Also, π^* not unique too

another $\pi =$
$$\begin{bmatrix} 0.4 & 0.1 & 0 \\ 0 & 0.3 & 0 \\ 0 & 0 & 0.2 \end{bmatrix}$$

⇒ Math:

- Support = $\{1, 2, 3\}$

- Cost [weight] = $C_{ij} = |1-j|$

↓

$$\begin{bmatrix} 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

- another cost matrix

$$(i-j)^2 =$$

- Cost is telling us if we have to
Probability mass from 1 to 3 will
be 2.

* Marginal constraints:

$$\Pi^* \cdot \mathbf{1} = P$$

$$\Pi^{*T} \cdot \mathbf{1} = Q$$

Π_{ij} = mass moved from $i \rightarrow j$ with
cost c_{ij}

not solvable by hand

$$\Rightarrow \sum_{i=1}^n \sum_{j=1}^n \pi_{ij} \quad (ij = \langle \pi, C \rangle)$$

$$\begin{aligned} \min \quad & \pi^* \langle \pi, C \rangle \text{ s.t.} \\ & 1 - \pi^* 1 = p \\ & 2 - \pi^* 1 = q \\ & 3, \pi_{ij} \geq 0 \end{aligned}$$

Sinkhorn-distance : a suboptimal
solvable

$$* \text{ Wasserstein distance} = W/D_1 = 0.1 * |1-2|$$

↓
based on L1 norm

$$W/D_2 = 0.1 * |1-2|^2$$

↓
based on L2 norm